XSuM as built (surrogate_official_m3_pfp, 2026-11-05): inputs, models, generated tables

one true neutrino interaction in, its reconstructed record out; tables are the XSuM cafpyana dataframes (mcnu, mcprim, mcdaught, slc, pfp)

INPUTS (detector-causal truth)

Final-state particle set [mcprim]

all post-FSI primaries, up to 32 tokens by KE
(0.45% of neutrinos lose their softest ones)
token: class (mu, e, p, n, pi+-, pi0, gamma, K+-, K0, Lambda, Sigma, other), px, py, pz, KE
+ path lengths from the vertex along its direction to the cathode, x/y wall, z wall

dropped: neutrinos, nuclei, Bertini pseudo-particle

80 summary features [mcnu, mcprim]

true vertex; distances to cathode and walls;
per species: multiplicity, summed / max KE, leading p, cos(theta), pT; totals (charged mult., visible E, EM energy, longitudinal momentum)

Truth not used as input

mcdaught (Geant4 daughters): copied, not an input
genie_mode, iscc, W, Q2, E_nu, signal flags: labels only

MODELS (trained on the training files only)

M3 particle-set encoder (stage A, then frozen)

transformer, 4 layers, d = 128, over particle tokens + a context token (80 features) -> 64-d embedding z; trained on topology, presence / containment flags and the slice block (3 M slices)

M2 Tier 0: what exists (input: 80 features + z)

matched slices per neutrino 0 / 1 / 2+: GBDT
topology: softmax MLP over 121 classes of (n_pfp, n_prong, n_trk, n_proton, n_shower), sampled first
GBDT heads: second track, two-track block, charge in 1-4 cm spheres, muon / pion-candidate containment flags, point masses (vertex at walls, back-to-back pairs)

M2 Tier 1: slice block (~50 numbers)

conditional flow matching, MLP 6 x 1024, masked per event to the groups that exist; conditioned on features, z, topology

M4 Tier 2: PFP set of each slice

softmax heads: number of fitted / unfitted PFPs
masked set flow (transformer, d = 256, 4 layers) over up to 12 PFP tokens x 29 fields; conditioned on features, z, topology and the generated slice block (309 numbers)

SURROGATED DATAFRAMES (production format)

slc_0: generated (M2)

vertex, nu_score, bcfm_score, total_pe, PFP counts;
longest / second track direction, length, PID, chi2;
two-track block (reco_t, opening angle, momenta, 12 end points, containment); charge in 1-4 cm spheres
pass_* flags recomputed from the generated values;
tmatch_idx = generating neutrino, tmatch_eff / purity NaN

pfp_0: generated (M4)

fitted PFPs only: trackScore, dist_to_vertex, primary flag, track start, direction, length, range / MCS p_mu, chi2 mu, p (2 calibrations), K, pi, PIDA (collection plane), shower energy, length, angle, density, gap, dE/dx, start
track end = start + length x direction (not generated)

copied from the production

mcnu, mcprim, mcdaught, hdr, POT tables

not in the surrogated files

cosmic-only slices and slices not matched to a neutrino
PFPs without a track or shower fit
induction-plane chi2, PFP truth-match columns

Validation done (held-out test files)

per-column / per-field marginals (W1), group presence and containment heads, cohpi cut flow (all and true-signal), correlation matrices and joint densities, per-event real-vs-generated correlations on the same neutrinos, CC1p0pi acceptance test (collaborator's cut flow, jobs 0-199): efficiency, purity and event overlap per stage

not done yet: physics holdouts, uncertainty estimates, end-to-end encoder, PFP-to-particle attention

Downstream (collaborators)

analyses run unchanged on the surrogated dataframes and compare with the real chain on the test files:
cohpi (S. Oh), kaon (T. Wester), CC1p0pi (M. Fernandez)

test-file loader: dataframe/xsum_testsplit.py

Training data and datasets (truth and reco linked by tmatch_idx)

SBND 2026 gen2 BNB + cosmics CV, v10_14_02_03: official SAM definition, 749,262 flat CAFs, 9.94 M events, processed as 2,000 grid jobs by a selection-free cafpyana configuration: 17.9 M true neutrinos, 5.74 M truth-matched slices. Split 80 / 10 / 10 by input file (never by row).
M2 and M4 trained on 3 M sampled training slices each (RTX 3090). Generated for all neutrinos: 5.92 M slices, 14.06 M PFPs.

Dataset: /pnfs/sbnd/persistent/users/lwan/dataframe/surrogate_official_m3_pfp/ (2,000 files, 60 GB; file_splits.parquet marks the test files).
Earlier: surrogate_official_v3 (slc only), surrogate_official_v3_pfp (no particle-set inputs; superseded for direction / containment cuts).